

Ch 11. Stabilization and State Estimation via Sliding Mode Control

Goal : Sliding mode control, Sliding mode observer for

linear time-invariant singularly perturbed systems subject to impulsive effects

$$(B) \begin{cases} \dot{x} = A_{11}x + A_{12}z + Bu & , \quad t \neq z_k \\ \varepsilon \dot{z} = A_{21}x + A_{22}z & , \quad t \neq z_k \\ x(t_k) = (I + E_k)x(t_k^-) & , \quad t = z_k \\ z(t_k) = (I + F_k)z(t_k^-) & , \quad t = z_k \\ x(0) = x_0, \quad z(0) = z_0 \end{cases}$$

$$\left(\begin{aligned} &x \in \mathbb{R}^n : \text{slow state}, \quad z \in \mathbb{R}^m : \text{fast state}, \quad u \in \mathbb{R}^r : \text{feedback control}, \\ &A_{11} \in \mathbb{R}^{n \times n}, \quad A_{12} \in \mathbb{R}^{n \times m}, \quad B \in \mathbb{R}^{n \times r}, \quad A_{21} \in \mathbb{R}^{m \times n}, \quad A_{22} \in \mathbb{R}^{m \times m}, \quad E_k \in \mathbb{R}^{n \times n}, \quad F_k \in \mathbb{R}^{m \times m}, \\ &z_k \nearrow \infty. \end{aligned} \right)$$

(*) : globally exponentially stabilized by u if $\|x(t)\| + \varepsilon \|z(t)\| \leq k(\|x_0\| + \varepsilon \|z_0\|)e^{-\lambda t}$

① Consider slow reduced, nonimpulsive subsystem.

(-1 later stabilize the whole system)

For $t \neq z_k$, setting $\varepsilon = 0$ yields

$$\begin{cases} \dot{x} = A_{11}x + A_{12}z + Bu \\ 0 = A_{21}x + A_{22}z \end{cases}$$

$$\Rightarrow \underline{z = h(x) = -A_{22}^{-1} A_{21} x} \quad (\text{We assume } A_{22} : \text{ nonsingular})$$

Reduced subsystem $\dot{\tilde{x}}_s = A_0 \tilde{x}_s + B_0 u_s, \quad A_0 = A_{11} - A_{12} A_{22}^{-1} A_{21}, \quad B_0 = B$
 $\in \mathbb{R}^{n \times n} \quad \in \mathbb{R}^{n \times m}$

• We consider r -dim sliding mode hyper-surface defined by

$$S_s(x_s) = \begin{pmatrix} s_1(x_s) \\ \vdots \\ s_r(x_s) \end{pmatrix} = \begin{pmatrix} c_1^T x_s \\ \vdots \\ c_r^T x_s \end{pmatrix} = \begin{pmatrix} c_1^T \\ \vdots \\ c_r^T \end{pmatrix} x_s = \underbrace{C_s}_{\mathbb{R}^{r \times n}} x_s$$

$$\Rightarrow \left(\frac{d}{dt} S_s(x_s) = \right) \dot{S}_s(x_s(t)) = C_s \dot{x}_s(t) = C_s A_0 x_s + C_s B_0 u_s$$

$$r\text{-dim equivalent control} : \underline{u_s^{eq} = -(C_s B_0)^{-1} C_s A_0 x_s} \quad \begin{matrix} \rightarrow \\ 0 \end{matrix}$$

$\rightarrow$ remains on the sliding surface $S_s(x_s) = 0$

• Analyze the motion outside the sliding surface, $S_s(x_s) \neq 0$

$$\text{Define } V(S_s(x_s)) = \frac{1}{2} S_s(x_s)^T S_s(x_s)$$

$$\text{Require } \dot{V}(S_s(x_s)) = S_s(x_s)^T \dot{S}_s(x_s) = S_s^T(x_s) (C_s A_0 x_s + C_s B_0 u_s) < 0,$$

$$\text{which is guaranteed by } u_s(t) = u_s^{eq}(t) - \underbrace{(C_s B_0)^T \text{diag}(\eta)}_{\text{rxr, positive diag els}} \widehat{\text{Sgn}(S_s(x_s))} = (\text{sgn}(s_{01}), \dots)$$

② Continuous closed-loop full system

$$\text{(outside sliding surface)} \begin{cases} \dot{x}_s = A_{11} x_s + A_{12} z + B u_s, & t \neq t_k \\ \dot{z} = A_{21} x_s + A_{22} z, & t \neq t_k \\ x_s(0) = x_{s0}, \quad z(0) = z_0 \end{cases}$$

(on the sliding surface) $\Sigma(x) > 0 \Rightarrow u_s(t) = u_s^{eq}(t)$ ($\Sigma(x_s(t)) = 0$)

$$\begin{cases} \dot{x}_s^* = A_{11}x_s + A_{12}z + B u_s^{eq} & t \neq t_k \\ \dot{z}^* = A_{21}x_s + A_{22}z & t \neq t_k \\ x_s(0) = x_{s0}, \quad z(0) = z_0 \end{cases}$$

Thm 11.1 Assume

① The reduced subsystem is stabilizable, A_{22} Hurwitz

② For $t \neq t_k$, $\exists a_{21}, a_{22} > 0$:

$$-2(z-h(x))^T P \dot{h}(x) \leq a_{21} x^T x + a_{22} (z-h(x))^T (z-h(x))$$

where P : m x m positive-definite, $A_{22}^T P + P A_{22} = -I$

③ $\exists \Sigma^* > 0$: $-\tilde{A}$: M-matrix (pg 52, Def 2.36)

$$\text{where } \tilde{A} = \begin{bmatrix} \frac{1}{\alpha_2} \max \{ \operatorname{Re}[\lambda(\tilde{A}_{11})] \} & \frac{1}{2\alpha_1 \beta_1} \\ \frac{a_{21}}{\alpha_1} & -(\frac{1}{\beta_2 \Sigma^*} - \frac{a_{22}}{\beta_1}) \end{bmatrix}$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2, \gamma_1 > 0$ defined later, $\tilde{A}_{11} = A_{11} + \frac{\gamma_1}{2} A_{12} A_{12}^T - A_{12} A_{12}^T A_{21} - B_1 (C B_1)^T C A_0$

④ For $i = 1, \dots, k$, $t_i - t_{i-1} > \frac{1}{v} \log(\alpha_{1i} + \alpha_{2i} + \beta_{2i})$, where

$v, \alpha_{1i}, \alpha_{2i}, \beta_{2i} > 0$ and $\alpha_{1i} + \alpha_{2i} + \beta_{2i} > 1$.

$\Rightarrow$ For each $\varepsilon \in (0, \Sigma^*]$, the full impulsive system becomes globally exp. stable.

pf) $x(t) = x(t; t_0, x_0, z_0)$, $z(t) = z(t; t_0, x_0, z_0)$: solution

$V(x) = \frac{1}{2} x^T x$, $W(z-h(x)) = (z-h(x))^T P (z-h(x))$: Lyapunov fn candidates for fast-slow subsystems.

For $\alpha_1 \leq \frac{1}{2}$, $\alpha_2 \geq \frac{1}{2}$, $\beta_1 \leq \lambda_{\min}(P)$, $\beta_2 \geq \lambda_{\max}(P)$,

$$\begin{cases} \alpha_1 \|x\|^2 \leq V(x) \leq \alpha_2 \|x\|^2, \\ \beta_1 \|z-h(x)\|^2 \leq W(z-h(x)) \leq \beta_2 \|z-h(x)\|^2 \end{cases}$$

time derivative of V along x during sliding motion ($\xi=0$)

$$\begin{aligned} \dot{V}(x) &= x^T \dot{x} = x^T (A_{11}x + A_{12}z + B_{1s}e_1^T) \\ &= x^T (A_{11}x + A_{12}(z-h(x)) + A_{12}h(x) - B_1(CB_0)^{-T}CA_0x) \\ &\leq x^T (A_{11} + \frac{\sigma_1}{2}A_{12}A_{12}^T - A_{12}A_{12}^TA_{11} - B_1(CB_0)^{-T}CA_0)x + \frac{1}{2\sigma_1}(z-h(x))^T(z-h(x)) \\ &\leq \frac{1}{\alpha_2} \max\{\rho_c[\lambda(\tilde{A}_{11})]\} V(x) + \frac{1}{2\sigma_1\beta_1} W(z-h(x)) \end{aligned}$$

time derivative of W along z during sliding motion

$$\dot{W}(z-h(x)) \leq -\left(\frac{1}{\beta_2\sigma_1} - \frac{q_{12}}{\beta_1}\right) W(z-h(x)) + \frac{q_{21}}{\alpha_1} V(x)$$

$$\Rightarrow \begin{pmatrix} \dot{V}(x) \\ \dot{W}(z-h(x)) \end{pmatrix} \leq \tilde{A} \begin{pmatrix} V(x) \\ W(z-h(x)) \end{pmatrix}, \quad \tilde{A}: M\text{-matrix by } \textcircled{2}$$

$$\Rightarrow \exists \xi > 0 : \forall t \in [z_k, z_{k+1}),$$

$$\begin{cases} V(x(t)) \leq \left(\|V(z_k)\| + \|W(z_k)\| \right) e^{-\xi(t-z_k)} \\ W(z-h(x)(t)) \leq \left(\|V(z_k)\| + \|W(z_k)\| \right) e^{-\xi(t-z_k)} \end{cases} \quad \textcircled{1}$$

$$(V(z_k) = V(x(z_k)), W(z_k) = W(z-h(x)(z_k)))$$

$$\text{At } t = z_k, \quad V(x(z_k)) \leq \alpha_{1k} V(x(z_k^-)), \quad \alpha_{1k} = \lambda_{\max}^2(I + E_k) \quad \textcircled{2}$$

$$W(z-h(x)(z_k)) = \dots \leq \beta_{2k} W(z-h(x)(z_k^-)) + \alpha_{2k} V(x(z_k^-)) \quad \textcircled{3}$$

$$t \in [t_0, z_1) \rightarrow \begin{cases} V(x(t)) \leq (||v(t_0)|| + ||w(t_0)||) e^{-\xi(t-t_0)} \\ W(z-L(x))|x| \leq \end{cases} \quad \square 1$$

$$t \in [z_1, z_2) \rightarrow \begin{cases} V(x(t)) \leq (||v(z_1)|| + ||w(z_1)||) e^{-\xi(t-z_1)} \\ \leq (\alpha_1 + \beta_{z_1} + \alpha_{z_1}) (||v(t_0)|| + ||w(t_0)||) e^{-\xi(z_1-t_0)} e^{-\xi(t-z_1)} \\ = (\alpha_1 + \beta_{z_1} + \alpha_{z_1}) (||v(t_0)|| + ||w(t_0)||) e^{-\xi(t-t_0)} \\ W(z-L(x))|x| \leq \end{cases} \quad \square 1 \quad \square 2, \square 3$$

$$\forall t \in [t_0, z_k) \rightarrow \begin{cases} V(x(t)) \leq \frac{k-1}{1} (\alpha_1 + \beta_{z_1} + \alpha_{z_1}) (||v(t_0)|| + ||w(t_0)||) e^{-\xi(t-t_0)} \\ W(z-L(x))|x| \leq \end{cases}$$

Choose $0 < \nu < \xi$. By ④, $\begin{cases} V(x(t)) \leq (||v(t_0)|| + ||w(t_0)||) e^{-(\xi-\nu)(t-t_0)} \\ W(z-L(x))|x| \leq \end{cases} \quad t \geq t_0$

$$\Rightarrow ||x(t)|| + ||z(t)|| \leq K (||x(t_0)|| + ||z(t_0)||) e^{-(\xi-\nu)(t-t_0)/2}, \quad t \geq t_0$$

(similar method in the pf of thm 9.1)

$$\begin{cases}
 \dot{x} = A_{11}x + A_{12}z + Bu & , \quad t \neq t_k \\
 \dot{z} = A_{21}x + A_{22}z & , \quad t \neq t_k \\
 x(t_k) = (I + E_k)x(t_k^-) & , \quad t = t_k \\
 z(t_k) = (I + F_k)z(t_k^-) & , \quad t = t_k \\
 x(0) = x_0, \quad z(0) = z_0
 \end{cases}$$

We can observe $y = Dx$, $D \in \mathbb{R}^{l \times n}$.

Design sliding mode observer through the reduced slow system.

Define the state estimate impulsive SPS :

$$\begin{cases}
 \dot{\hat{x}} = A_{11}\hat{x} + A_{12}\hat{z} + Bu + Lv(\hat{y} - y) & , \quad t \neq t_k \\
 \dot{\hat{z}} = A_{21}\hat{x} + A_{22}\hat{z} & , \quad t \neq t_k \\
 \hat{x}(t_k) = (I + E_k)\hat{x}(t_k^-) & , \quad t = t_k \\
 \hat{z}(t_k) = (I + F_k)\hat{z}(t_k^-) & , \quad t = t_k \\
 \hat{x}(0) = \hat{x}_0, \quad \hat{z}(0) = \hat{z}_0
 \end{cases}$$

L : observer gain matrix, v : r-dim observer to be designed.
 $\uparrow$
 $\mathbb{R}^{n \times r}$

error states $e_x = \hat{x} - x$, $e_z = \hat{z} - z$ satisfy

$$\begin{cases}
 \dot{e}_x = A_{11}e_x + A_{12}e_z + L\bar{v}(e_x) & , \quad t \neq t_k \\
 \dot{e}_z = A_{21}e_x + A_{22}e_z & , \quad t \neq t_k \\
 e_x(t_k) = (I + E_k)e_x(t_k^-) & , \quad t = t_k \\
 e_z(t_k) = (I + F_k)e_z(t_k^-) & , \quad t = t_k
 \end{cases}$$

$$|e_{x0}| = e_{x0}, \quad |e_{z0}| = e_{z0}$$

$$\bar{v}(e_x) = v(|De_x|)$$

Setting $v=0$,

$$\boxed{\text{reduced error subsystem}} \quad \dot{e}_{x_s} = A_e e_{x_s} + L_s \bar{v}(e_x),$$

$$\text{where } e_z = h(e_x) = -A_{12}^T A_{22} e_x, \quad A_e = A_{11} - A_{12} A_{22}^T A_{21}, \quad L_s = L$$

$$\text{Sliding mode error surface} : S_e(e_{x_s}) = C_e e_{x_s}, \quad C_e \in \mathbb{R}^{r \times n}$$

$$\bar{v}_{eq}(e_{x_s}) = -(C_e L_s)^T C_e A_e e_{x_s}$$

$$e_{x_s}^* = (I - L_s (C_e L_s)^T C_e) A_e e_{x_s} =: A_e^{eq} e_{x_s}$$

$$V(S_e(e_x)) = \frac{1}{2} S_e(e_{x_s})^T S_e(e_{x_s})$$

$$\rightarrow \dot{V}(S_e) < 0 \text{ if } \bar{v}(e_{x_s}) = \bar{v}_{eq}(e_{x_s}) - (C_e L_s)^T \text{diag}(\eta) \text{sgn}(S_e(e_{x_s}))$$

sliding mode control law

Thm 11.2 Assume

① reduced slow, fast subsystems are observable

$$\textcircled{2} \exists \varepsilon^* > 0 : -\tilde{A} : M\text{-matrix}, \quad \tilde{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & -(\frac{1}{\varepsilon^*} - a_{22}) \end{bmatrix}$$

$$\textcircled{3} t_i - t_{i-1} > \frac{1}{\gamma} \log(\alpha_{e_i} + \alpha_{z_i} + \beta_{z_i}), \quad \gamma, \alpha_{e_i}, \alpha_{z_i}, \beta_{z_i} > 0, \quad \alpha_{e_i} + \alpha_{z_i} + \beta_{z_i} > 1$$

$\Rightarrow$ closed-loop full-order error system : globally exponentially stable, $\forall \varepsilon \in (0, \varepsilon^*)$

pf) Similar to thm 11.1